

Customer Churn Analysis

Business Analyst Project Report

1. Executive Summary

This project analyzes telecom customer churn to understand why customers leave, identify high-risk segments, and recommend retention actions. The project uses Excel, SQL, Power BI, Python and Machine Learning.

The dataset contains 7,043 customers. 1,869 customers churned, giving an overall churn rate of 27%.

2. Business Problem

Business Question: Why are customers leaving, which customers are most at risk, and what should the company do to reduce churn?

3. KPI Summary

Metric	Value
Total Customers	7,043
Churned Customers	1,869
Retained Customers	5,174
Overall Churn Rate	27%

4. Key Findings

Dimension	Segment	Churn Rate
Contract	Month-to-month	43%
Contract	One year	11%
Contract	Two year	3%
Tenure	0–12 months	47%
Tenure	13–24 months	29%
Tenure	25–48 months	32%
Tenure	49+ months	13%
Internet	DSL	19%
Internet	Fiber optic	42%
Internet	No internet	7%
Payment	Electronic check	45%
Payment	Mailed check	19%
Payment	Bank transfer	17%
Payment	Credit card	15%
Tech Support	No	42%
Tech Support	Yes	15%
Device Protection	No	39%
Device Protection	Yes	23%
Online Security	No	42%

Dimension	Segment	Churn Rate
Online Security	Yes	15%
Monthly Charges	High	35%
Monthly Charges	Medium	24%
Monthly Charges	Low	12%

5. Contract × Tenure

Month-to-month customers show the highest churn across tenure groups: 51% for 0–12 months, 38% for 13–24 months, 33% for 25–48 months and 26% for 49+ months. Longer contracts have substantially lower churn.

6. High-Risk Segment

Segment	Customers	Churned	Churn Rate
All Customers	7,043	1,869	27%
High-Risk Conditions	754	534	71%

7. Business Recommendations

Finding	Recommended Action
Month-to-month churn = 43%	Encourage migration to one-year/two-year contracts with suitable incentives.
0–12 month churn = 47%	Create stronger onboarding and first-year engagement programs.
Fiber optic churn = 42%	Investigate pricing, service quality, installation and customer experience.
Electronic check churn = 45%	Promote automatic payment methods.
No Tech Support churn = 42%	Offer proactive technical support and support bundles.
No Online Security churn = 42%	Promote security bundles and communicate their value.
High monthly charges churn = 35%	Use targeted plan reviews and retention offers.

8. Tools Used

Excel: cleaning and KPI analysis. **SQL:** aggregation and business questions. **Power BI:** interactive dashboard. **Python:** analysis and ML. **Machine Learning:** Logistic Regression churn prediction.

9. Conclusion

Churn is concentrated among month-to-month customers, newer customers, fiber-optic customers, electronic-check users, customers without support/security services and customers with higher monthly charges. These groups should receive targeted retention attention.